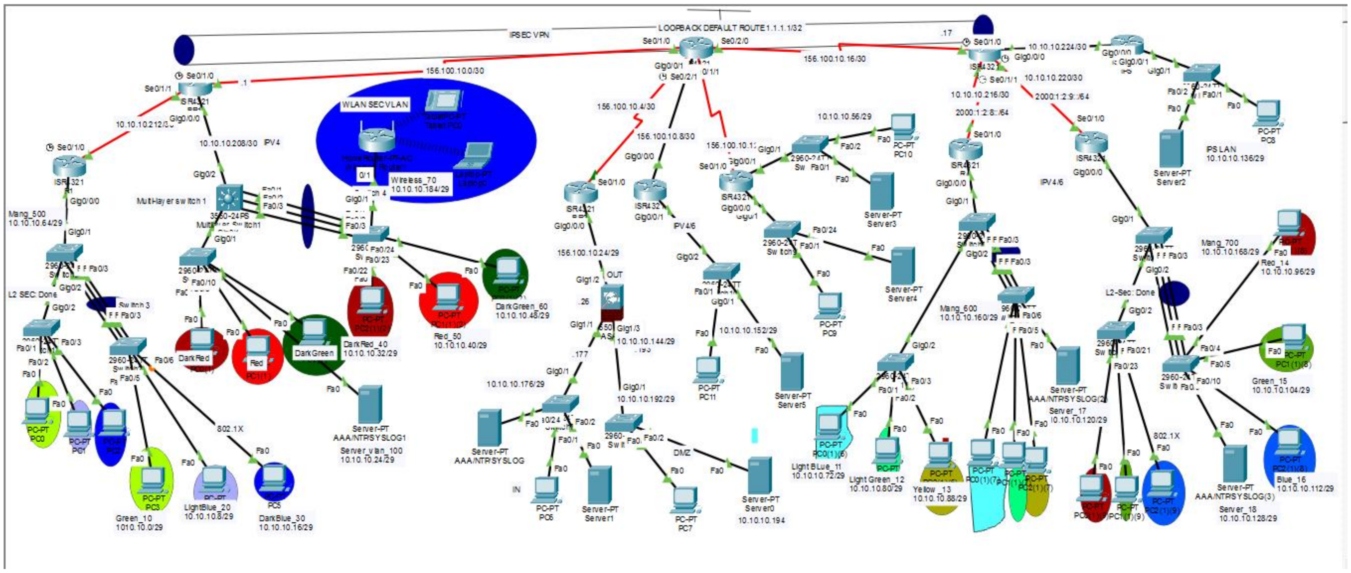
Design and Implementation of a Secure Three-Branch Enterprise Network



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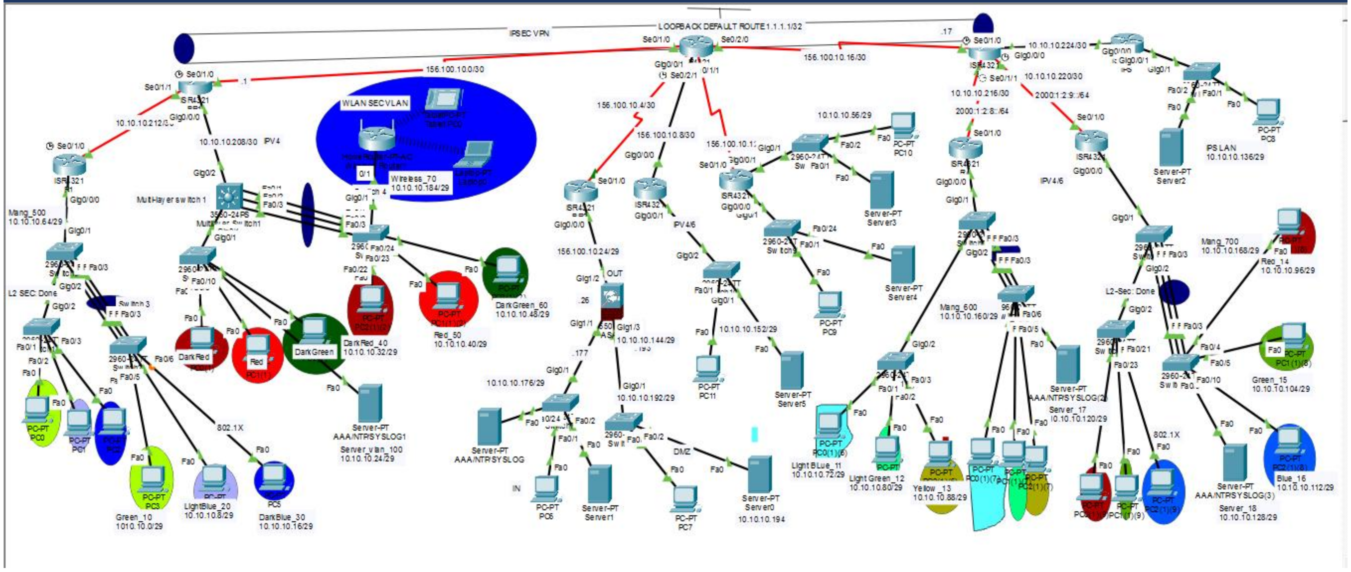
NTI Network Project — Comprehensive Documentation

Part 1: Global Project Standards



The infrastructure follows a hierarchical design (Core, Distribution, Access) and adheres to the following administrative, connectivity, and security standards:

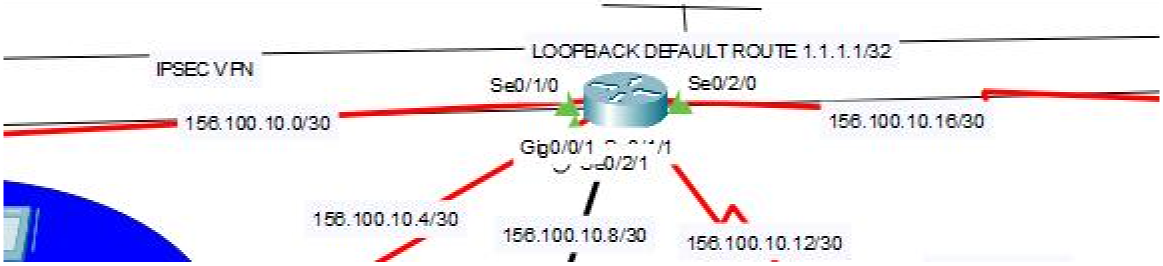
- Routing Protocol: OSPFv2 (Process 1, Area 0) for IPv4 dynamic routing across all branches and the ISP. MD5 authentication is enabled on all OSPF interfaces using key ismail.
- Dual-Stack: Stateful DHCPv6 is implemented on the ACL Router (Middle Branch) for the 2000:1:2:1::/64 prefix. IPv6 P-T-P links use the 2000:1:2:x::/64 addressing scheme.
- Switching & Encapsulation: IEEE 802.1Q (Dot1q) for all trunk links and sub-interfaces (Router-on-a-Stick inter-VLAN routing).
- EtherChannel: LACP (active mode) for link aggregation between core and access switches.
- Management: SSH Version 2 with domain cisco.com. Telnet is disabled. VTY exec-timeout: 5 minutes. SSH retries: 3.
- VLAN Security: Unused ports → VLAN 999 (Black Hole), administratively shutdown, DTP disabled, CDP disabled. Native VLAN on all trunks → VLAN 777 (prevents VLAN hopping).
- AAA: TACACS+ for administrative device access on Routers. RADIUS for user 802.1X endpoint authentication and access switches.
- L2 Security: Port Security (sticky MAC, max 1, violation shutdown), DHCP Snooping (trusted uplinks only), Dynamic ARP Inspection (DAI).
- VPN: Site-to-site IPSec using AES-256, SHA, DH Group 5, ESP. Tunnel: BR1 ↔ BR3.
- Firewalls & IPS: Cisco ASA and ZPF on the middle Branch, IOS IPS inline on the IPS router on Branch 3.

Basic configuration (standard Template):

```
hostname <Based on device>

enable secret ciscopriv

banner motd "Authorized Access Only!"

ip domain-name ismail.local
```

```
crypto key generate rsa modulus 1024

username admin secret cisco

line console 0
 login authentication default
exit

ip ssh version 2
line vty 0 4
 transport input ssh
 login authentication default
exit

service password-encryption

do wr
```

Part 2: ISP Router



The ISP Router (Router-ID: 10.10.10.10) acts as the WAN hub, providing connectivity between all branch border routers using public /30 point-to-point links in the 156.100.10.0/24 space. All interfaces participate in OSPF Area 0.

1. WAN Interface Addressing

✔ All WAN IPs corrected from 203.0.113.x/30 placeholders to the actual 156.100.10.x/30 addressing scheme.

Interface	Connection To	IPv4 Address	IPv6 Address	OSPF MD5 Key
`Se0/1/0`	Branch 1 (BR1)	`156.100.10.2/30`	—	`ismail`
`Se0/1/1`	Branch 2 Edge (BR2)	`156.100.10.6/30`	—	`ismail`
`Gi0/0/1`	ACL Router (Middle)	`156.100.10.10/30`	`2000:1:2:a::2/64`	`ismail`
`Se0/2/1`	ZPF Router (Right)	`156.100.10.14/30`	—	`ismail`
`Se0/2/0`	BR3 (Right Border)	`156.100.10.18/30`	`2000:1:2:b::1/64`	`ismail`

2. ISP Routing & Services

- OSPFv2 (Process 1, Router-ID 10.10.10.10): Manages IPv4 reachability across the WAN. Advertises a default route via default-information originate to all connected branches.
- OSPF MD5 Authentication: Key ismail (key ID 1) enabled on all five WAN interfaces.
- IPv6 P-T-P Links: 2000:1:2:a::/64 (ISP ↔ ACL Router) and 2000:1:2:b::/64 (ISP ↔ BR3).

Part 2: Consolidated Project Documentation (Final)

1. Global Network Standards

- Management: SSH Version 2, domain cisco.com. Loopback 0 (10.255.255.1/32) as management source on applicable routers.
- Centralized AAA: TACACS+ server at 10.10.10.26 (Left), 10.10.10.122 (Right/R4), 10.10.10.130 (Right/R5). RADIUS server at 10.10.10.122.
- Monitoring: NTP primary → 10.10.10.122 (Server 1, VLAN 17). NTP secondary → 10.10.10.130 (Server 2, VLAN 18). Syslog primary → 10.10.10.122.
- VLAN Security: Black-hole VLAN 999. Native VLAN 777 (replaces the placeholder VLAN 99 used in the draft).

2. Branch 1 (Internal & Edge)

- Core/Distribution: R1 handles inter-VLAN routing (Router-on-a-Stick) for VLANs 10/20/30 on Gi0/0/0 sub-interfaces. L3-MLS handles VLANs 40/50/60/100.
- DHCP: R1 provides DHCP for VLANs 10 (Green_Dept), 20 (LightBlue_Dept), 30 (DarkBlue_Dept). MLS provides DHCP for VLANs 40/50/60/100.
- Edge (BR1): Terminates Site-to-Site IPSec VPN to peer 156.100.10.17 (ZPF Router / BR3 side).

3. Global VLAN Strategy

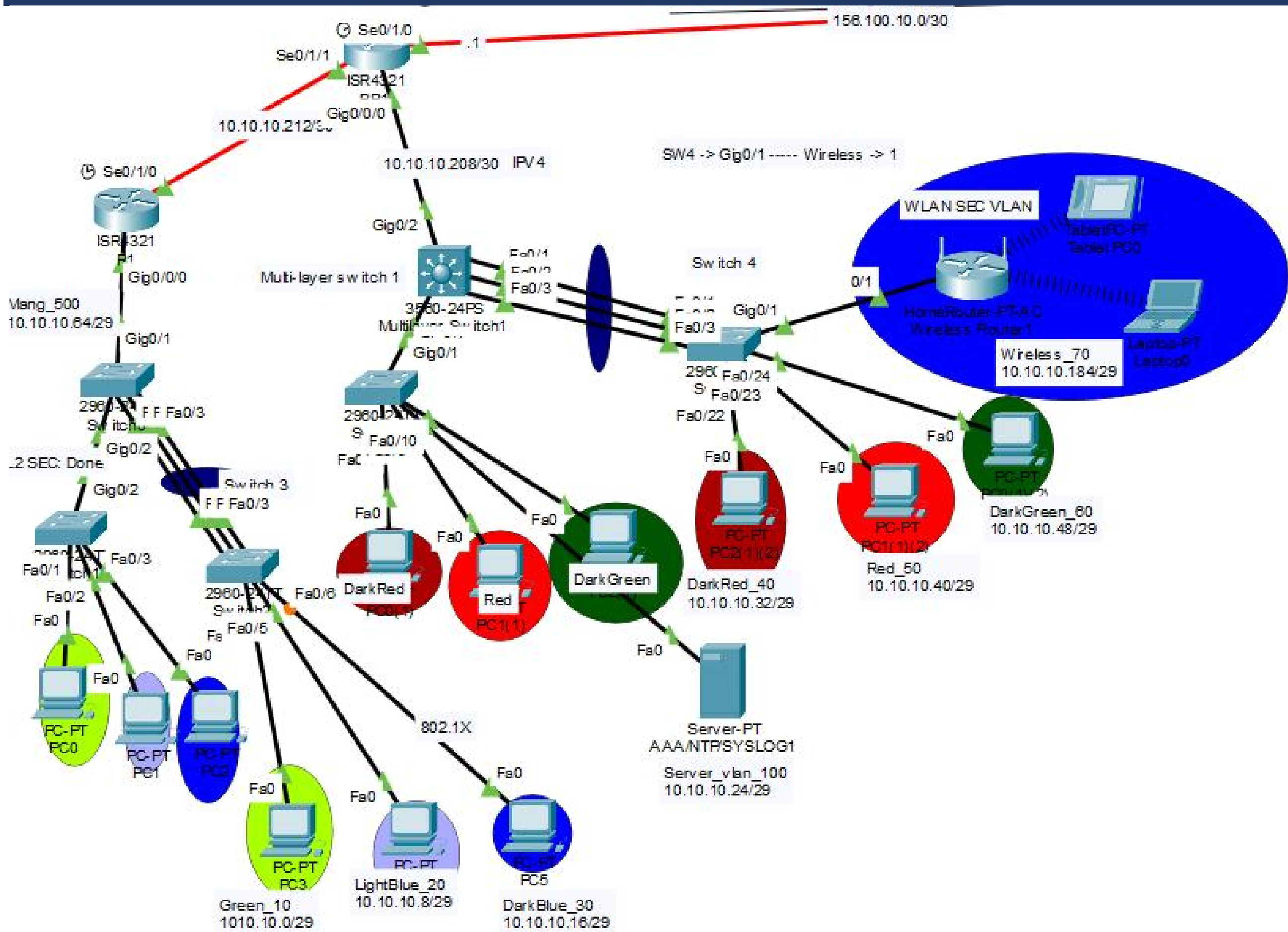
VLAN ID	Name	Subnet	Purpose
10	**Green_Dept**	`10.10.10.0/29`	Left Branch User Data — R1 Gi0/0/0.10
20	**LightBlue_Dept**	`10.10.10.8/29`	Left Branch User Data — R1 Gi0/0/0.20
30	**DarkBlue_Dept**	`10.10.10.16/29`	Left Branch User Data — R1 Gi0/0/0.30
40	**DarkRed_Dept**	`10.10.10.32/29`	Left Branch / MLS
50	**Red_Dept**	`10.10.10.40/29`	Left Branch / MLS
60	**DarkGreen_Dept**	`10.10.10.48/29`	Left Branch / MLS
70	**Wireless_Router**	`10.10.10.184/29`	Left Branch Wireless
100	**Server_VLAN**	`10.10.10.24/29`	Left Branch Servers
11	**Light Blue**	`10.10.10.72/29`	Right Branch — R4 Gi0/0/0.11
12	**Light Green**	`10.10.10.80/29`	Right Branch — R4 Gi0/0/0.12
13	**Yellow**	`10.10.10.88/29`	Right Branch — R4 Gi0/0/0.13
14	**Red**	`10.10.10.96/29`	Right Branch — R5 Gi0/0/0.14
15	**Green**	`10.10.10.104/29`	Right Branch — R5 Gi0/0/0.15
16	**Blue**	`10.10.10.112/29`	Right Branch — R5 Gi0/0/0.16
17	**Server 1**	`10.10.10.120/29`	Right Branch — R4 Gi0/0/0.17 | Mgmt Server
18	**Server 2**	`10.10.10.128/29`	Right Branch — R5 Gi0/0/0.18 | Backup Server

777	**Native_VLAN**	N/A	Trunk security — untagged native traffic (replaces VLAN 99)
999	**Blackhole**	N/A	Security — all unused ports assigned, DTP off, CDP off, shutdown

4. Infrastructure Connectivity (P2P Links)

Segment	Link Subnet	IPv6	Key Protocols
BR1 ↔ MLS (Left Internal)	10.10.10.208/30	—	OSPFv2 Area 0, MD5 ismail
BR1 ↔ R1 (Left Internal)	10.10.10.212/30	—	OSPFv2 Area 0, MD5 ismail
R4 ↔ BR3 (Right)	10.10.10.216/30	2000:1:2:8::/64	OSPFv2 Area 0, MD5 ismail
R5 ↔ BR3 (Right)	10.10.10.220/30	2000:1:2:9::/64	OSPFv2 Area 0, MD5 ismail
IPS ↔ BR3 (Right)	10.10.10.224/30	—	OSPFv2 Area 0, MD5 ismail
BR1 ↔ ISP (WAN)	156.100.10.0/30	—	OSPFv2, MD5 ismail
BR2 ↔ ISP (WAN)	156.100.10.4/30	—	OSPFv2, MD5 ismail
ACL Router ↔ ISP (WAN)	156.100.10.8/30	2000:1:2:a::/64	OSPFv2 + Dual-Stack, MD5 ismail
ZPF ↔ ISP (WAN)	156.100.10.12/30	—	OSPFv2 + IPSec VPN map
BR3 ↔ ISP (WAN)	156.100.10.16/30	2000:1:2:b::/64	OSPFv2, MD5 ismail
ASA ↔ ISP (WAN)	156.100.10.24/29	—	Static route / NAT

Branch 1: Internal Infrastructure



A. Addressing Table (IPv4)

✔ VLANs corrected: 10/11/12 → 10/20/30. L3-SW VLANs corrected: 13-16 → 40/50/60/100. All IPs verified against /29 subnet scheme.

Device	Interface	IP Address	Role
R1	`Gi0/0/0.10`	`10.10.10.1/29`	VLAN 10 — Green_Dept GW
	`Gi0/0/0.20`	`10.10.10.9/29`	VLAN 20 — LightBlue_Dept GW
	`Gi0/0/0.30`	`10.10.10.17/29`	VLAN 30 — DarkBlue_Dept GW
MLS (L3-SW)	`Vlan 40`	`10.10.10.33/29`	VLAN 40 — DarkRed_Dept SVI GW
	`Vlan 50`	`10.10.10.41/29`	VLAN 50 — Red_Dept SVI GW
	`Vlan 60`	`10.10.10.49/29`	VLAN 60 — DarkGreen_Dept SVI GW
	`Vlan 70`	`10.10.10.185/29`	VLAN 70 — Wireless GW
	`Vlan 100`	`10.10.10.25/29`	VLAN 100 — Server_VLAN SVI

			GW
	Gi0/2	10.10.10.209/30	P-T-P → BR1
BR1	Se0/1/0	156.100.10.1/30	WAN → ISP (OSPF + VPN)
	Se0/1/1	10.10.10.214/30	BR1 ↔ R1
	Gi0/0/0	10.10.10.210/30	BR1 ↔ MLS

B. Connectivity & Trunking

Service	Component	Details & Configuration
Trunking & EtherChannel — S0 ↔ S2	Port-channel1 — FA0/1, FA0/2, FA0/3	LACP (Active Mode) | Native: 777 | Allowed: 10, 20, 30
Trunking & EtherChannel — MLS ↔ S4	Port-channel1 — FA0/1, FA0/2, FA0/3	LACP (Active Mode) | Native: 777 | Allowed: 40, 50, 60, 70, 100
OSPFv2 Area 0	Connectivity: BR3, R1 and MLS	MD5 Authentication (Key: hosny)
	Passive Interfaces (R1)	`Gi0/0/0.10, .20, .30`
	Passive Interfaces (MLS)	`Vlan40, 50, 60, 70, 100`
DHCP (R1)	VLAN 10	`Pool: 10.10.10.0/29 | GW: 10.10.10.1 | DNS: 8.8.8.8`
	VLAN 20	`Pool: 10.10.10.8/29 | GW: 10.10.10.9 | DNS: 8.8.8.8`
	VLAN 30	`Pool: 10.10.10.16/29 | GW: 10.10.10.17 | DNS: 8.8.8.8`
DHCP (MLS)	VLAN 40	`Pool: 10.10.10.32/29`
	VLAN 50	`Pool: 10.10.10.40/29`
	VLAN 60	`Pool: 10.10.10.48/29`
	VLAN 70	`Pool: 10.10.10.184/29`

C. L2 Security (Left Branch Switches)

Switch	Feature	Interfaces	Snooped VLANs	Detail
S0 (Core)	VLAN Hardening	`Fa0/4-24 (blackhole)`	Native: 777	Trunks: allow 10,20,30,500,777
S1	Port Security	`Fa0/1(V10), Fa0/2(V20), Fa0/3(V30)`	—	Sticky MAC, max 1, violation shutdown
	DHCP Snooping	`Gi0/2 (trust)`	VLANs 10, 20, 30	Rate limit Fa0/1-3 → 5 pps, option82 off
S2	Port Security	`Fa0/4(V10), Fa0/5(V20),`	—	Sticky MAC, max 1, violation shutdown

		`Fa0/6(V30)`		
S0 (Core)	DAI	`Gi0/1, Gi0/2 (trust)`	VLANs 10, 20, 30	`ip arp inspection vlan 10,20,30`

Monitoring and AAA Servers — Branch 1

Service	Protocol	Server IP	Key / Details	Applicable Devices
Monitoring	**Syslog**	`10.10.10.26`	Centralized Logging (SIEM-ready)	All Network Devices
	NTP	`10.10.10.26`	Time Sync (msec precision)	All Network Devices
AAA	**TACACS+**	`10.10.10.26`	`Key: mohamed123`	BR1, R1, MLS
	RADIUS	`10.10.10.26`	`Key: h123456`	All Switches

D. Edge Layer & WAN (BR1 Router)

WAN Connectivity

Interface	IP Address	Connected To	Notes
`Se0/1/0`	`156.100.10.1/30`	ISP Se0/1/0	OSPFv2 + IPSec crypto map applied here
`Se0/1/1`	`10.10.10.213/30`	R1 (internal P-T-P)	MD5 OSPF auth, network 10.10.10.212/30
`Gi0/0/0`	`10.10.10.209/30`	MLS (internal P-T-P)	MD5 OSPF auth, network 10.10.10.208/30

Site-to-Site VPN (IPSec) — BR1

Parameter	Value
VPN Peer	`156.100.10.17 (ZPF Router / BR3 WAN)`
Phase 1 — Encryption	`aes 256`
Phase 1 — Hash	`sha`
Phase 1 — DH Group	`group 5`
Phase 1 — Auth	`pre-share  (PSK: cisco)`
Phase 1 — Lifetime	`3600 seconds`
Phase 2 — Transform-Set	`branch1: esp-aes 256 esp-sha-hmac`
Phase 2 — SA Lifetime	`1000 seconds`
PFS Group	`group5`
Crypto Map	`vpn_crypto seq 10 — applied on Se0/1/0`

Interesting Traffic ACL	vpn_traffic ACL: BR1 subnets (0.0.0.0–.208) ↔ BR3 subnets (.72–.224)

NTP & Syslog Configuration (BR1)

Server: 10.10.10.26 (Left Branch Management Server)

```
ntp server 10.10.10.26
service timestamps log datetime msec
logging host 10.10.10.26
logging on
```

TACACS+ Configuration (BR1, R1, MLS)

```
aaa new-model
tacacs-server host 10.10.10.26
tacacs-server key ismail123
aaa authentication login default group tacacs+ local
line console 0
 login authentication default
exit
line vty 0 4
 login authentication default
exit
```

RADIUS Configuration (All Switches — Branch 1)

```
aaa new-model
radius-server host 10.10.10.26 auth-port 1645
radius-server key h123456
aaa authentication login default group radius local
line console 0
 login authentication default
exit
line vty 0 4
 login authentication default
exit
```

AAA Server Accounts

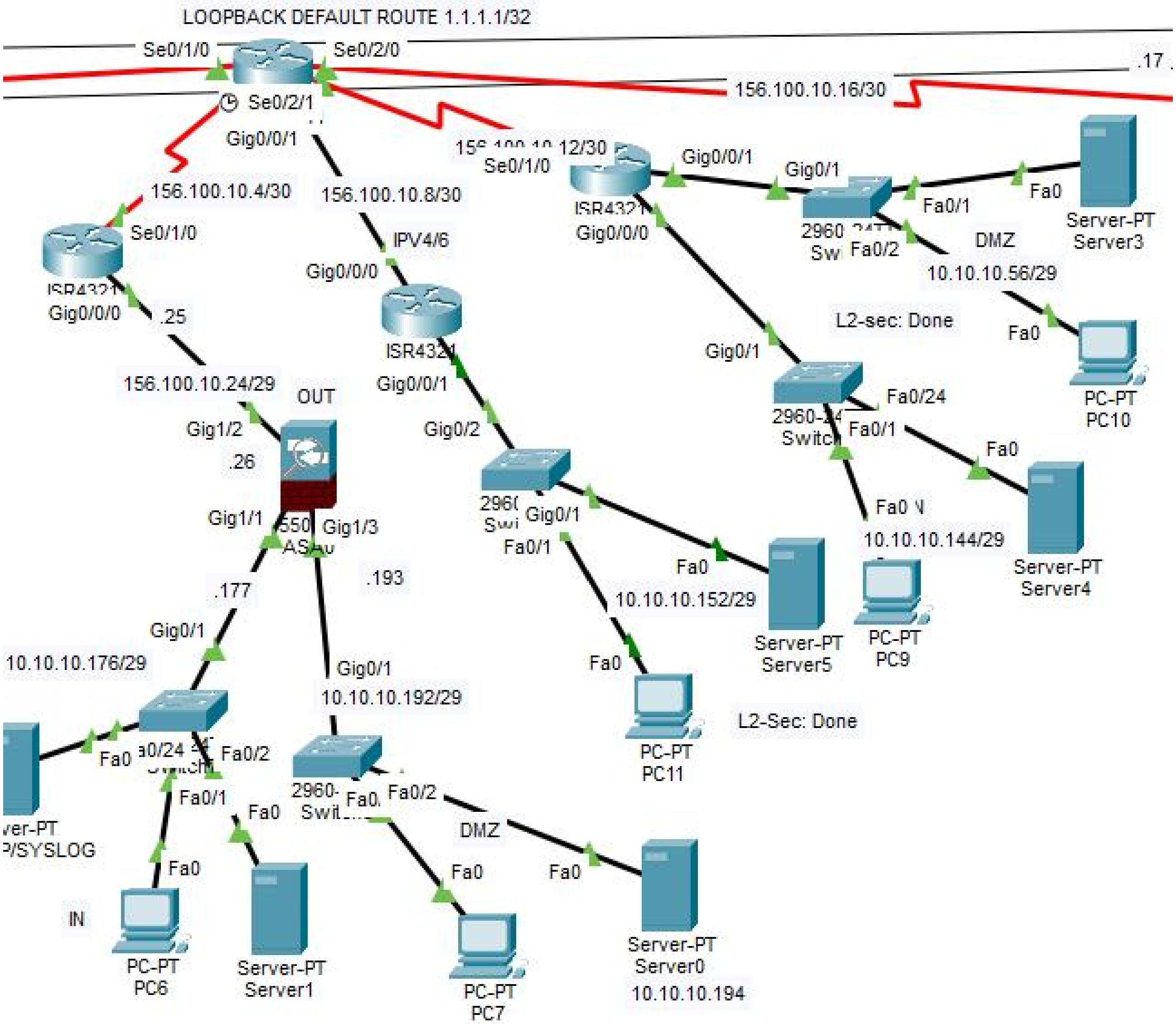
Username	Password
admin	cisco
Ismail	Ismail123

Branch 1 — Summary Table

Component	Detail
Devices	BR1 (edge), R1 (inter-VLAN), MLS (L3-SW), S0 (Core), S1, S2 (access), S3, S4 (MLS access)
VLANs Served	VLANs 10 (Green), 20 (LightBlue), 30 (DarkBlue) via R1 | VLANs 40, 50, 60, 70, 100 via MLS
Inter-VLAN Routing	Router-on-a-Stick (R1 Gi0/0/0 sub-interfaces) + MLS SVI routing
EtherChannel	Port-channel1 (LACP Active) — S0↔S2 (VLANs 10-30) and MLS↔S4 (VLANs 40-100)
WAN Link	BR1 Se0/1/0 → 156.100.10.1/30 (ISP). OSPF Area 0, MD5 key:

	`ismail`
VPN Tunnel	`BR1 ↔ ZPF (BR3) | Peer: 156.100.10.17 | AES-256, SHA, DH-5, ESP`
DHCP	R1: VLANs 10/20/30 | MLS: VLANs 40/50/60/70 | DNS: 8.8.8.8
L2 Security	Port Security (sticky, max 1, shutdown) | DHCP Snooping | DAI | VLAN 999 blackhole | Native 777
AAA & Monitoring	`TACACS+/RADIUS/Syslog/NTP → 10.10.10.26 | Key: Ismail123 (TACACS), h123456 (RADIUS)`
OSPF Passive	`R1: Gi0/0/0.10/.20/.30 | MLS: Vlan40/50/60/70/100`
Admin Access	`SSH v2 | domain: Ismail.local | user: admin | secret: ciscopriv | key RSA 1024`

Branch 2 & Perimeter Security



1. Adaptive Security Appliance (ASA) Configuration

Interface Security Levels

Interface	Nameif	IP Address	Sec Level	Purpose
`GigabitEthernet1/1`	**INSIDE**	`10.10.10.177/29`	**100**	Internal LAN
`GigabitEthernet1/2`	**OUTSIDE**	`156.100.10.26/29`	**0**	WAN (→ ISP router 156.100.10.25)
`GigabitEthernet1/3`	**DMZ**	`10.10.10.193/29`	**50**	DMZ Server Segment

NAT Implementation

Object	NAT Type	Translation	Effect
INSIDE-NET (10.10.10.176/29)	Dynamic NAT	→ OUTSIDE interface IP	All inside hosts share 156.100.10.26 for internet
DMZ-SERVER (10.10.10.194)	Static NAT	`→ 156.100.10.27`	DMZ server publicly reachable on fixed public IP

- Access Control: OUTSIDE-DMZ ACL permits external HTTP/HTTPS to 10.10.10.194 (DMZ server). Applied inbound on OUTSIDE interface.
- Service Policy: global_policy inspects ICMP globally (stateful ICMP tracking).
- Routing: Static default route → 156.100.10.25. OSPF Process 1 for internal reachability.
- SSH Management: RSA 2048-bit. Allowed from 10.10.10.178/32 (INSIDE) and 10.10.10.3/32 (OUTSIDE). Timeout: 10 min. User: hosny / Pass: cisco.

2. ACL & Dual-Stack Router (ACL Router / Middle Branch)

Addressing & DHCP

Interface	IPv4 Address	IPv6 Address	Purpose
`Gi0/0/1`	`10.10.10.153/29`	`2000:1:2:1::1/64`	LAN — DHCPv4 pool acl-r-ipv4 + DHCPv6 stateful (managed-flag)
`Gi0/0/0`	`156.100.10.9/30`	`2000:1:2:A::1/64`	WAN → ISP (OSPF Area 0 + MD5 auth. Key = ismail)

Access Control Lists (Applied inbound on Gi0/0/0)

#	Action	Protocol	Source	Destination / Port	Purpose
1	**permit**	`OSPF`	`156.100.10.8/30`	any	OSPF hellos from ISP link
2	**permit**	`TCP`	any	`10.10.10.155 :80`	HTTP to internal server
3	**permit**	`TCP`	any	`10.10.10.155 :443`	HTTPS to internal server
4	**permit**	`ICMP echo-reply`	any	`10.10.10.152/29`	Return ping only — no unsolicited ICMP
5	**deny**	all	any	any	Implicit deny

3. Zone-Based Policy Firewall Router (ZPF — Right Branch Border)

Zone Definitions & Interface Mapping

Zone	Interface	IP Address	DHCP Pool
Inside	`Gi0/0/0`	`10.10.10.145/29`	zpf-left: network 10.10.10.144/29, GW .145
DMZ	`Gi0/0/1`	`10.10.10.57/29`	zpf-right: network 10.10.10.56/29, GW .57

Outside	Se0/1/0	156.100.10.13/30	WAN → ISP (OSPF Area 0 + MD5 auth. Key = ismail)

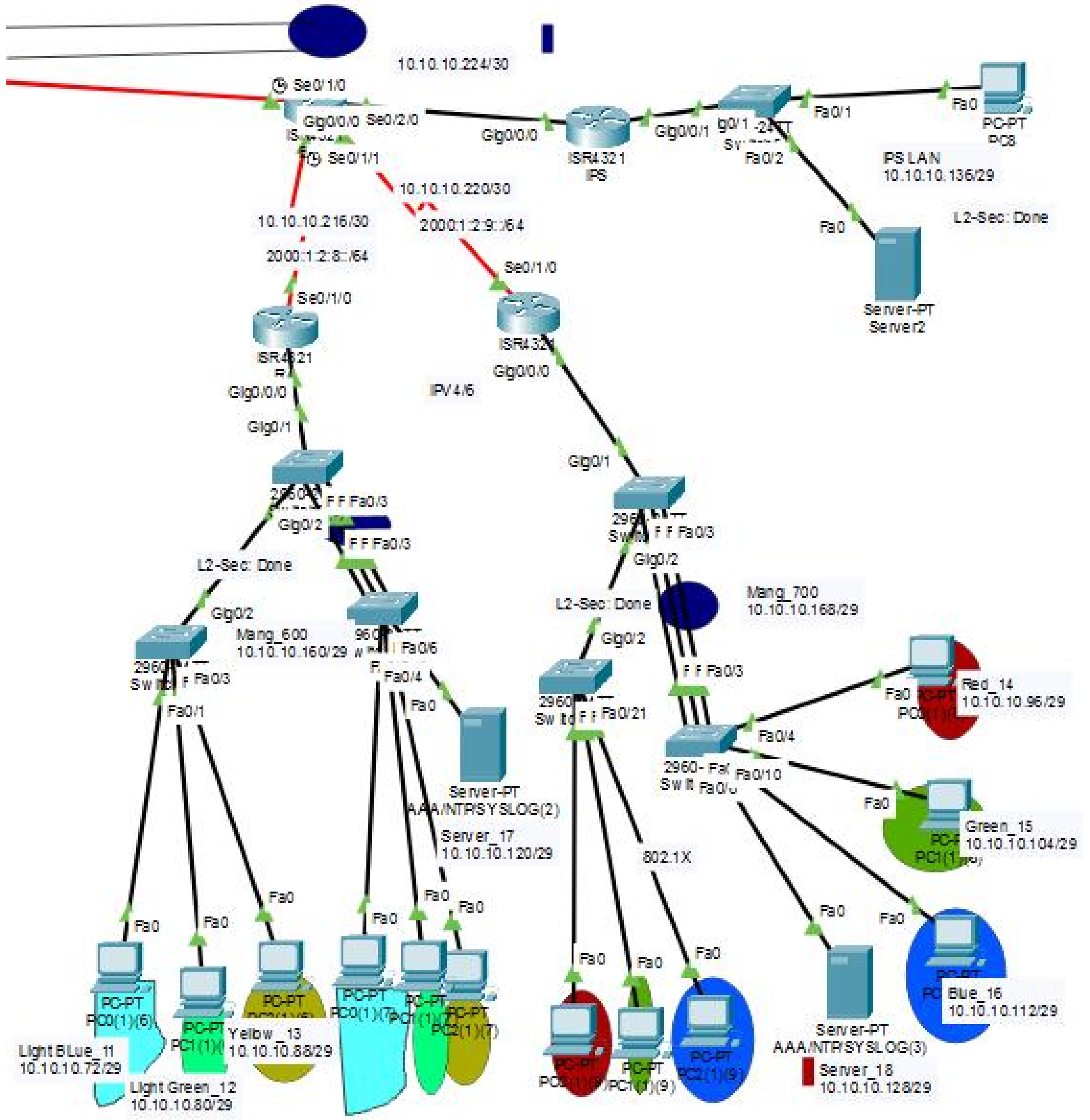
Security Policies

Source Zone	Dest Zone	Verdict	Protocols Inspected	Policy-Map
Inside	Outside	✓ **Allow**	TCP, UDP, ICMP	in-to-out-policy
Inside	DMZ	✓ **Allow**	ICMP only	in-to-dmz-policy
Outside	Inside	✗ **Deny**	All — default drop	class-default drop
Outside	DMZ	✓ **Allow**	HTTP, HTTPS	out-to-dmz-policy
DMZ	Inside	✗ **Deny**	All — default drop	class-default drop
DMZ	Outside	✓ **Allow**	ICMP only	dmz-to-out-policy

Branch 2 — Summary Table

Component	Detail
Devices	BR2 (edge), ACL-46-R (dual-stack), ZPF-R (firewall), ASA (hardware FW), S6 (DMZ switch), S9 (inside switch), S10 (ACL switch)
ASA Interfaces	INSIDE: 10.10.10.177/29 (SL100) | OUTSIDE: 156.100.10.26/29 (SL0) | DMZ: 10.10.10.193/29 (SL50)
ASA NAT	Dynamic PAT: INSIDE-NET → OUTSIDE interface | Static NAT: 10.10.10.194 → 156.100.10.27
ZPF Zones	Inside (Gi0/0/0): 10.10.10.145/29 | DMZ (Gi0/0/1): 10.10.10.57/29 | Outside (Se0/1/0): 156.100.10.13/30
ZPF Policy	In→Out: TCP+UDP+ICMP (inspect) | In→DMZ: ICMP only | Out→DMZ: HTTP+HTTPS | DMZ/Out→In: DROP
ACL Router	LAN: 10.10.10.153/29 + 2000:1:2:1::1/64 | WAN: 156.100.10.9/30 + 2000:1:2:A::1/64
ACL Rules	Permit: OSPF from WAN, TCP/80 & TCP/443 to .155, ICMP echo-reply to .152/29 | Deny: all else
BR2 WAN	Se0/1/0: 156.100.10.5/30 → ISP | OSPFv2 Area 0, MD5 key: ismail
Admin Access	SSH v2 | domain: ismail.local | user: admin | secret: ciscopriv | key RSA 1024

Branch 3 Infrastructure



A. Addressing Tables

WAN and Core Connectivity

Device	Interface	IPv4 Address	IPv6 Address	Connection / Role
BR3 (border)	`Se0/1/0`	`156.100.10.17/30`	`2000:1:2:B::2/64`	WAN → ISP (IPSec VPN map applied)
	`Se0/1/1 (→R4)`	`10.10.10.218/30`	`2000:1:2:8::2/64`	P-T-P → R4
	`Se0/2/0 (→R5)`	`10.10.10.222/30`	`2000:1:2:9::2/64`	P-T-P → R5
	`Gi0/0/0 (IPS LAN)`	`10.10.10.225/30`	—	IPS LAN segment gateway

IPS Router	`Gi0/0/1`	`10.10.10.138/29`	—	IPS LAN (inline inspection)
IPS ↔ BR3	`Gi0/0/0`	`10.10.10.226/30`	—	P-T-P → BR3
R4 ↔ BR3	`Se0/1/0`	`10.10.10.217/30`	`2000:1:2:8::1/64`	P-T-P → BR3
R5 ↔ BR3	`Se0/1/0`	`10.10.10.221/30`	`2000:1:2:9::1/64`	P-T-P → BR3

Internal Departmental Gateway — R4 (Router-on-a-Stick)

Sub-Interface	IPv4 Address	IPv6 Address	VLAN	Department / Role
`Gi0/0/0.11`	`10.10.10.73/29`	`2000:1:2:2::1/64`	**11**	Light Blue (User Data)
`Gi0/0/0.12`	`10.10.10.81/29`	`2000:1:2:3::1/64`	**12**	Light Green (User Data)
`Gi0/0/0.13`	`10.10.10.89/29`	`2000:1:2:4::1/64`	**13**	Yellow (User Data)
`Gi0/0/0.17`	`10.10.10.121/29`	`2000:1:2:C::1/64`	**17**	Server 1 — Management (NTP/Syslog/AAA)

Internal Departmental Gateway — R5 (Router-on-a-Stick)

Sub-Interface	IPv4 Address	IPv6 Address	VLAN	Department / Role
`Gi0/0/0.14`	`10.10.10.97/29`	`2000:1:2:5::1/64`	**14**	Red (User Data)
`Gi0/0/0.15`	`10.10.10.105/29`	`2000:1:2:6::1/64`	**15**	Green (User Data)
`Gi0/0/0.16`	`10.10.10.113/29`	`2000:1:2:7::1/64`	**16**	Blue (User Data)
`Gi0/0/0.18`	`10.10.10.129/29`	`2000:1:2:D::1/64`	**18**	Server 2 — Backup Management (NTP/Syslog/AAA)

B. Connectivity & Trunking

Service	Component / Device	Details & Configuration
Trunking & EtherChannel	Port-channel1 — FA0/1, FA0/2, FA0/3	LACP (Active Mode) | Native: 777 | Allowed: 11,12,13,14,15,16,17,18,777
OSPFv2 Area 0	Connectivity: BR3, R4, R5, IPS Router	MD5 Authentication (Key: ismail)
	Passive Interfaces (R4)	`Gi0/0/0.11, .12, .13, .17, .600`
	Passive Interfaces (R5)	`Gi0/0/0.14, .15, .16, .18, .700`
DHCP (R4)	VLAN 11	`Pool: 10.10.10.72/29 | GW: 10.10.10.73 | DNS: 8.8.8.8`
	VLAN 12	`Pool: 10.10.10.80/29 | GW: 10.10.10.81 | DNS: 8.8.8.8`
	VLAN 13	`Pool: 10.10.10.88/29 | GW: 10.10.10.89`

		`| DNS: 8.8.8.8`
DHCP (R5)	VLAN 14	`Pool: 10.10.10.96/29 | GW: 10.10.10.97 | DNS: 8.8.8.8`
	VLAN 15	`Pool: 10.10.10.104/29 | GW: 10.10.10.105 | DNS: 8.8.8.8`
	VLAN 16	`Pool: 10.10.10.112/29 | GW: 10.10.10.113 | DNS: 8.8.8.8`

C. L2 Security (Branch 3 Switches)

Switch	Feature	Interfaces	Snooped VLANs	Detail
S0(3) Core	VLAN Hardening	`Fa0/4-24 (blackhole)`	Native: 777	Trunks: allow 11,12,13,17,600,777
	DAI	`Gi0/1, Gi0/2 (trust)`	VLANs 11, 12, 13	`ip arp inspection vlan 11,12,13`
S0(4) Core	VLAN Hardening	`Fa0/4-24 (blackhole)`	Native: 777	Trunks: allow 14,15,16,18,700,777
	DAI	`Gi0/1, Gi0/2 (trust)`	VLANs 14, 15, 16	`ip arp inspection vlan 14,15,16`
S1(3)	Port Security	`Fa0/1(V11), Fa0/2(V12), Fa0/3(V13)`	—	Sticky MAC, max 1, violation shutdown
	DHCP Snooping	`Gi0/2 (trust)`	VLANs 11, 12, 13	Rate limit Fa0/1-3 → 5 pps, option82 off
S2(3)	Port Security	`Fa0/4(V11), Fa0/5(V12), Fa0/6(V13)`	—	Sticky MAC, max 1, violation shutdown
S1(4)	Port Security	`Fa0/21(V16), Fa0/22(V15), Fa0/23(V14)`	—	Sticky MAC, max 1, violation shutdown
	DHCP Snooping	`Gi0/2 (trust)`	VLANs 14, 15, 16	Rate limit Fa0/21-23 → 5 pps, option82 off
S2(4)	Port Security	`Fa0/4(V14), Fa0/5(V15), Fa0/6(V16)`	—	Sticky MAC, max 1, violation shutdown

D. Edge Layer & WAN (BR3 Router)

WAN Connectivity

Interface	IP Address	Connected To	Notes
`Se0/1/0`	`156.100.10.17/30`	ISP Se0/2/0	OSPFv2 + IPSec crypto map applied here
`Se0/1/1`	`10.10.10.218/30`	R4 (internal P-T-P)	MD5 OSPF auth (Key: ismail), network 10.10.10.216/30

Se0/2/0	10.10.10.222/30	R5 (internal P-T-P)	MD5 OSPF auth (Key: ismail), network 10.10.10.220/30
Gi0/0/0	10.10.10.225/30	IPS (internal P-T-P)	MD5 OSPF auth (Key: ismail), network 10.10.10.224/30

Monitoring and AAA Servers — Branch 3

Service	Protocol	Server IP	Key / Details	Applicable Devices
Monitoring	**Syslog**	10.10.10.122	Centralized Logging (SIEM-ready)	All Network Devices
		10.10.10.130	Backup	All Network Devices
	NTP	10.10.10.122	Time Sync (msec precision)	All Network Devices
		10.10.10.130	Backup	All Network Devices
AAA	**TACACS+**	10.10.10.122	Key: ismail123	BR3, R4, R5
		10.10.10.130	Backup — Key: ismail123	BR3, R4, R5
	RADIUS	10.10.10.122	Key: ismail123	All Switches
		10.10.10.130	Backup — Key: ismail123	All Switches

AAA Server Accounts

Username	Password
admin	cisco
ismail	ismail123

Site-to-Site VPN (IPSec) — BR3

Parameter	Value
VPN Peer	156.100.10.1 (BR1 WAN)
Phase 1 — Encryption	aes 256
Phase 1 — Hash	sha
Phase 1 — DH Group	group 5
Phase 1 — Auth	pre-share (PSK: cisco)
Phase 1 — Lifetime	3600 seconds
Phase 2 — Transform-Set	branch3: esp-aes 256 esp-sha-hmac

Phase 2 — SA Lifetime	1000 seconds
PFS Group	group5
Crypto Map	vpn_crypto seq 10 — applied on Se0/1/0
Interesting Traffic ACL	vpn_traffic ACL: BR3 subnets (.72-.224) ↔ BR1 subnets (0.0.0.0-.208)

Branch 3 — Summary Table

Component	Detail
Devices	BR3 (edge/VPN), R4 (inter-VLAN), R5 (inter-VLAN), IPS-BR3 (inline IPS), S0(3)/S0(4) (core), S1-S2 per stack (access), S5-IPS
VLANs — R4	VLAN 11 (10.10.10.72/29) | 12 (10.10.10.80/29) | 13 (10.10.10.88/29) | 17 Server1 (10.10.10.120/29)
VLANs — R5	VLAN 14 (10.10.10.96/29) | 15 (10.10.10.104/29) | 16 (10.10.10.112/29) | 18 Server2 (10.10.10.128/29)
EtherChannel	Port-channel1 (LACP Active) on both switch stacks (3) and (4)
WAN Link	BR3 Se0/1/0 → 156.100.10.17/30 (ISP) | OSPFv2 Area 0, MD5 key: ismail
VPN Tunnel	BR3 ↔ BR1 | Peer: 156.100.10.1 | AES-256, SHA, DH-5, ESP | Transform-set: branch3
IPS	IOS IPS inline on IPS-BR3. Category ios_ips basic. Sig 2004: deny-packet-inline (drops echo-reply to server). Alerts → 10.10.10.139
AAA & Monitoring	TACACS+ → 10.10.10.122 (key ismail123) | RADIUS → 10.10.10.122 | NTP+Syslog → 10.10.10.122 (primary) / .130 (backup)
IPv6	Dual-stack on R4/R5 sub-interfaces. P-T-P: 2000:1:2:8::/64 (R4↔BR3), 2000:1:2:9::/64 (R5↔BR3)
L2 Security	Port Security (sticky, max 1, shutdown) | DHCP Snooping (VLANs 11-16) | DAI | VLAN 999 blackhole | Native 777
Admin Access	SSH v2 | domain: ismail.local | user: admin | secret: ciscopriv | key RSA 1024